

Uvažujme reálnou matici $A = \begin{pmatrix} 1 & 1 & 1 & -2 \\ 1 & 1 & -2 & 1 \\ 1 & -2 & 1 & 1 \end{pmatrix}$

(10.1) Spočítejte singulární rozklady matic A , A^T , AA^T a $A^T A$.

Porobíme nejprve singulární rozklad matice $A^T = A^*$.

Dle věty o singulárním rozkladu hledáme

- ON báze vektorů $\mathbb{R}^3, \mathbb{R}^4$; tedy

$$B = (\vec{v}_1, \vec{v}_2, \vec{v}_3)$$

$$C = (\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4)$$

- singulární hodnoty $0 < \sigma_1, \sigma_2, \sigma_3 \in \mathbb{R}$
(rank $A^T = 3$);

které, že pro $D = [d_{ij}]_{4 \times 3}^B = \text{diag}_{4 \times 3}(\sigma_1, \sigma_2, \sigma_3)$,

$$U = [id]_{4 \times 4}^C = (\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4),$$

$$V = [id]_{3 \times 3}^B = (\vec{v}_1, \vec{v}_2, \vec{v}_3)$$

(tj. U, V jsou ortogonální) platí $A^T = UDV^T = UDV^*$.

Víme, že singulární hodnoty jsou druhé odmocniny vl. čísel matice $(A^T)^* A^T = AA^T$ a $\vec{v}_1, \vec{v}_2, \vec{v}_3$ jsou vlastní vektory příslušné těmto vlastním číslům.

Eigensystem[a . at]

{{9, 9, 3}, {{-1, 0, 1}, {-1, 1, 0}, {1, 1, 1}}}

$$\Rightarrow \boxed{\sigma_1 = 3; \sigma_2 = 3; \sigma_3 = \sqrt{3}}$$

Orthogonalize[{{-1, 0, 1}, {-1, 1, 0}}]

$$\boxed{\left\{ \left\{ -\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}} \right\}, \left\{ -\frac{1}{\sqrt{6}}, \sqrt{\frac{2}{3}}, -\frac{1}{\sqrt{6}} \right\} \right\} = (\vec{v}_1, \vec{v}_2)}$$

$$\boxed{\vec{v}_3 = \frac{1}{\sqrt{3}} (1, 1, 1)^T}$$

$$V^* = V^T = \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A_{4 \times 3}^T = U_{4 \times 4} D_{4 \times 3} V_{3 \times 3}^T$$

$$/ \cdot (V^T)^{-1} = (V^T)^T = V$$

$$A^T V = U D$$

V je ortogonální

$$(A^T \vec{v}_1 | A^T \vec{v}_2 | A^T \vec{v}_3) = (U \vec{d}_1 | U \vec{d}_2 | U \vec{d}_3)$$

$$(A^T \vec{v}_1 | A^T \vec{v}_2 | A^T \vec{v}_3) = (\sqrt{3} \vec{u}_1 | \sqrt{6} \vec{u}_2 | \sqrt{3} \vec{u}_3)$$

$$\Downarrow$$

$$\vec{u}_1 = \frac{1}{\sqrt{3}} A^T \vec{v}_1 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & -2 \\ -2 & 1 & 1 \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \frac{1}{\sqrt{3}} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{pmatrix}$$

$$\vec{u}_2 = \frac{1}{\sqrt{6}} A^T \vec{v}_2 = \left(0, \frac{1}{\sqrt{6}}, -\frac{2}{\sqrt{6}}, \frac{1}{\sqrt{6}}\right)^T$$

$$\vec{u}_3 = \frac{1}{\sqrt{3}} A^T \vec{v}_3 = (1, \dots, \dots)^T$$

Doplníme na ON bázi prostoru $\mathbb{R}^4$:

$$U_{4 \times 3}' \sim \begin{pmatrix} 0 & 0 & 1 & -1 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

$$\vec{u}_4' = (0, 1, 1, 1)^T \in \text{Ker } U'$$

$$\vec{u}_4 = \frac{\vec{u}_4'}{\|\vec{u}_4'\|} = \frac{1}{\sqrt{3}} (0, 1, 1, 1)^T$$

$$A^T = U D V^T$$

$$A^T = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & -\sqrt{\frac{2}{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & \sqrt{3} \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

Ověřte pro komplexní matice platí:

$$(A^T)^* = (U D V^T)^*$$

$$A = V D^* U^* = V D^T U^T$$

Jelikož známe OG matice V, U a diag. D , známe i singulární rozklad pro matice A :

$$A = V D^T U^T$$

$$A = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \sqrt{\frac{2}{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & \sqrt{3} & 0 \end{pmatrix} \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{6}} & -\sqrt{\frac{2}{3}} & \frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

Snadno přijde i na singul. rozkladz matic AA^* , A^*A .

$U, V \dots$ ortogonální

$$AA^* = AA^T = VD^T U^T U D V^T \stackrel{\downarrow}{=} VD^T D V^T = V \cdot \text{diag}_{3 \times 3}(9, 9, 3) \cdot V^T$$

$$A^*A = A^T A = U D V^T V D^T U^T = U D D^T U^T = U \cdot \text{diag}_{4 \times 4}(9, 9, 3) \cdot U^T$$

$$AA^* = V \cdot \text{diag}_{3 \times 3}(9, 9, 3) \cdot V^T$$

$$AA^* = \begin{pmatrix} -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & \sqrt{\frac{2}{3}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 3 \end{pmatrix} \cdot \begin{pmatrix} -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{6}} & \sqrt{\frac{2}{3}} & -\frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$

$$A^*A = U \cdot \text{diag}_{4 \times 4}(9, 9, 3) \cdot U^T$$

$$A^*A = \begin{pmatrix} 0 & 0 & 1 & 0 \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ 0 & -\sqrt{\frac{2}{3}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \end{pmatrix} \cdot \begin{pmatrix} 9 & 0 & 0 & 0 \\ 0 & 9 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \cdot \begin{pmatrix} 0 & -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{6}} & -\sqrt{\frac{2}{3}} & \frac{1}{\sqrt{6}} \\ 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{pmatrix}$$